

SPREADSHEET REVIEW, MEASURED IN MONEY

Gridlint

Your board deck says 38.6 months of runway.

It has 5.2.

Gridlint reads the formulas in a workbook, recalculates every cell, and finds the mistakes that quietly change the answer — ranked by how much money each one moves.

Yusuke Ota · University of Tokyo · solo entry

github.com/taro13nyanko/gridlint

THE PROBLEM

A broken spreadsheet does not look broken.

A SUM that stops one row short still returns a number. It still formats as currency. It still lands in the board deck.

Field audits find errors in 24% to 94% of real business spreadsheets. Panko found at least one in 94% of 88 audited workbooks. (EuSpRIG)

=SUM(C11:C14)

Row 15 is "Contractors", \$185,000 a month.

The total is right there in the deck, and it is short every single month.

What people use today

PerfectXL · Operis OAK · Spreadsheet Detective

Desktop add-ins, \$249–\$2,000 per year.
They hand an analyst 400 warnings.

Excel's own error checking

Green triangles for a handful of patterns.
No ranking, no impact, no fix.

A second pair of eyes

The most common control in practice, and the least reliable one.

None of them tells you which warning is worth reading, or what the number becomes once it is fixed.

THE SOLUTION

Rank by money. Then prove the fix.

1

Read the formulas

Every formula is parsed into a syntax tree and linked into a dependency graph.

2

Recalculate the workbook

Gridlint's own engine recomputes every cell — then checks itself against the values Excel saved.

3

Price each defect

The fix is applied to a copy, the workbook is recomputed, and the difference is measured.

4

Say it in plain English

A model writes the note. It never decides what is wrong and never produces a number.

The output is not a list of warnings. It is a sentence with a number in it:

“One SUM leaves out the Contractors row. Runway goes from 38.6 months to 5.2, and gross margin from +6.3% to –30%.”

WHO IT IS FOR

Whoever signs off on the number.

Finance and ops teams

Month-end close, board packs, budget re-forecasts. The ritual is monthly and the file is the same one every time.

Analysts and consultants

They inherit a model built by somebody who left. Gridlint is the first hour of that handover.

Grant and programme admins

Public money, spreadsheet reporting, an auditor at the end of it.

Pricing

Free

\$0

One workbook · every rule

Team

\$29 / editor / month

Unlimited workbooks · history · shareable reports · corrected file

Close

\$99 / month

Scheduled re-checks · CI check for committed workbooks

It is a subscription because the mistake comes back every month, with the next version of the file.

One screen: what is wrong, and what it costs.

 Gridlint

How it works Proof Pricing Sign in

← Check another file

✓ engine matches the file on 123/123 values 3 critical 3 warning

board-model.xlsx

123 formulas across 2 sheets · checked in 126 ms

BIGGEST MEASURED IMPACT

SUM covers C11:C14 but "Contractors" sits right below it and is left out.

Cost per head \$175,859 → \$243,659 Estimated tax provision \$2,640 → \$0 Runway (months) 38.6 → 5.2 Gross margin 6.3% → -30%

Findings

critical 3 warning 3 info 0

● Total leaves out a row of data

Operating Model!C16 \$2,779,774

12 cells · 30 cells change · R001

● One formula differs from the rest of the line

Operating Model!K13 \$183,141

16 cells change · R002

● IFERROR is masking #DIV/0!

Operating Model!C29 R012

● A number is stored as text inside C32:C34

Operating Model!C35 \$38,500 R008

● Hidden assumption: 0.0825

Operating Model!C28 R003

● Hidden assumption: 1.35

Operating Model!K13 R003

Total leaves out a row of data

Operating Model!C16 and 11 more · rule R001 · confidence 97%

SUM covers C11:C14 but "Contractors" sits right below it and is left out. The same mistake is repeated in 12 cells (Operating Model!C16 to N16).

FORMULA

=SUM(C11:C14)

↓ Extend C11:C14 to C11:C15 in all 12 cells (Operating Model!C16 to N16)

=SUM(C11:C15)

NUMBERS PEOPLE READ THAT CHANGE

LINE	NOW	AFTER THE FIX
Cost per head	\$175,859	\$243,659
Estimated tax provision	\$2,640	\$0
Runway (months)	38.6	5.2
Gross margin	6.3%	-30%

✓ fix verified by recalculation

The fix was applied to a copy, the workbook recomputed, and 30 cells changed. No new errors appeared.

Ranked by measured impact

The biggest number is the one that moves the most money, not the first one found.

Grouped

Twelve monthly totals with the same mistake are one finding, not twelve.

Verified fix

“30 cells change, no new errors” — because it was actually run.

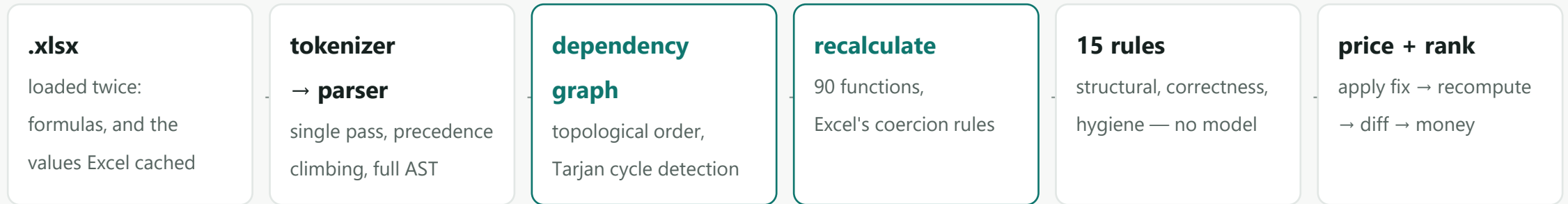
The grid, flagged

Toggle formulas and the defect is visible in place.

Gridlint · AI Builders Hackathon 2026

5

An Excel engine, written from scratch.



The self-check

Every workbook already stores the values its app last computed. Gridlint recomputes them and compares.

That single comparison does two jobs: it validates the engine, and it detects a file saved with calculation switched off — the numbers on screen no longer matching the formulas underneath.

Below 99.5% agreement, Gridlint says so and withholds the money figures rather than guessing.

Engineering decisions

Evaluation never recurses into other cells: the graph is sorted, then evaluated in order. A 3,000-cell chain cannot blow the stack, and there is a test for it.

Ranges are clamped to the used area, so =SUM(A:A) does not expand to a million nodes.

Impacts are never summed — they overlap. An odd mistake for this product to make.

The model writes. The code decides.

Parse formulas, build the graph	Code
Recalculate the workbook	Code
Decide whether something is a defect	Code
Measure what a fix changes	Code
Explain the defect to a non-expert	Model, fenced
Draft a repair where no mechanical fix exists	Model, executed

Two fences, both tested

1 • Number guard

Every figure in a written note must already appear in the evidence the detector measured. Anything else is a number the model invented, and the whole sentence is thrown away.

2 • Executed repairs

A proposed formula is parsed, run through the engine, and diffed. If it introduces one new error cell anywhere, it is rejected and never shown.

Delete the model entirely and Gridlint still finds every defect and still measures every impact. The model makes the report readable; it does not make it true.

Four numbers, all reproducible from the repository.

156 / 156

formulas where the engine matches Excel exactly

A conformance workbook generated by Excel itself: $-2^2 = 4$, $2^3^2 = 64$, ROUND half away from zero, text ignored in a range but coerced as an argument.

182 / 182

planted defects found by the right rule

Eight kinds of defect injected into twelve clean workbooks, every mutant recalculated by Excel, then scored on whether the right rule named the right cell.

0

findings on 12 clean workbooks, 1,059 formulas

Blank spacer rows, subtotals rolled into a grand total, cross-sheet assumptions, VLOOKUP tables, hidden helper columns. A checker that cries wolf gets uninstalled.

The benchmark earned its keep.

It caught a real bug in Gridlint's own arithmetic: ROUND used a relative epsilon to correct binary representation error, which turned 3,806,241.4967 into 3,806,242. It now rounds in decimal.

And a fourth number: 7 seconds.

A 23,000-formula model, audited end to end. Profiling it found two quadratic passes that had been costing 37 seconds. The evaluator runs at 88,000 formulas a second.

The check moves from “someone should look at this” to something that runs.

Before a number leaves the building

Drop the file in, read one sentence, decide. The whole check takes under a second on a 123-formula model.

Every month, on the same file

A workspace keeps the workbook and its history, so the second month is a re-check, not a re-review.

In CI, on a workbook in a repository

`gridlint check model.xlsx --fail-on critical` exits non-zero. A pull request that breaks a total fails the build, the way it would for code.

As a shareable report

A read-only link is how a reviewer says “here is what I found” to whoever owns the file, without sending the file back.

What is built, and what is next.

Working today

- Formula engine, 90 functions, 156/156 against Excel
- Dependency graph, recalculation, cycle detection
- 15 rules, impact measured by recalculation
- Web app: workspaces, history, shared reports, corrected download
- Dependency trace: where a number actually comes from
- CLI, CI workflow, Docker, 151 tests

Next

- Google Sheets: read a file directly instead of exporting
- More of Excel: array formulas, dynamic arrays, LET and LAMBDA
- Rule packs per domain (grant reporting, payroll, VAT)
- Watch mode: re-check on every save, diff against last month
- A public benchmark corpus so other tools can be measured against it

I pay for university by fixing other people's spreadsheets. Every single one had a silent formula bug in it.

github.com/taro13nyanko/gridlint · MIT